AERO5560 Group 8
Aircraft Modelling and Pitch-Rate Feedback
Progress Report: Data, Trim, Linearisation and Inner-Loop Architecture

Group 8

19 October 2026

Abstract

This report documents the completed extraction of Group 8 aircraft data, trim verification, numerical linearisation, and the elevator-to-pitch-rate transfer function, together with the construction and labelling of the pitch-rate feedback loop required by the assignment. The supplied initial condition, approximately 200 knots at 300 ft, satisfies numerical equilibrium and maintains its initial speed and attitude during a 60 s undisturbed nonlinear simulation. Five central-difference step sizes are used to obtain the full 12×12 state matrix A and 12×5 input matrix B, followed by a fixed-altitude, four-state longitudinal model. The resulting $G_{q\delta_e}(s)$ has a zero at the origin and a ZPK gain of approximately -88.196939. The feedback architecture uses unity measurement and an ideal elevator actuator, with all command, error, control and output signals defined. Closed-loop and sensitivity–complementary-sensitivity–compensator (SCK) expressions are derived. The compensator $K_q(s)$ remains to be designed; no claim is made that the closed-loop performance specifications have been met.

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1 Scope and Modelling Conventions

This stage addresses two requirements in Longitudinal Task 1: determining $G_{q\delta_e}$ for the assigned aircraft, and constructing a fully labelled feedback loop that controls pitch rate using the elevator. The preceding data extraction, trim verification and linearisation provide the model used for these requirements. The three compensator designs, outer loop, gust analysis, auto-throttle and complete autopilot integration remain subsequent tasks.

The full state vector contains body-axis velocities and angular rates, Euler attitude angles, and earth-frame positions:

$$\boldsymbol{X} = [u, v, w, p, q, r, \phi, \theta, \psi, X_e, Y_e, Z_e]^{\mathsf{T}}. \tag{1}$$

The control-input vector is

$$\boldsymbol{U} = [\delta_T, \delta_e, \delta_a, \delta_r, \delta_f]^{\mathsf{T}}, \tag{2}$$

representing throttle, elevator, aileron, rudder and flap, respectively. Internal calculations use SI units: m/s for velocity, m for position, rad for angles, rad/s for angular rates, and a 0–1 throttle fraction. Altitude is $h = -Z_e$, and vertical speed is $V_s = -\dot{Z}_e$, positive upwards.

Linear-model variables are increments about the nominal flight trajectory:

$$\boldsymbol{X} = \boldsymbol{X}_{\text{ref}}(t) + \Delta\boldsymbol{X}, \qquad \boldsymbol{U} = \boldsymbol{U}_0 + \Delta\boldsymbol{U}. \tag{3}$$

Because the aircraft continues to move forward in earth coordinates, the X_e component of $\boldsymbol{X}_{\text{ref}}(t)$ is not constant. The position derivatives must not all be required to vanish, and an incremental control deflection must not be treated as an absolute deflection.

2 Group 8 Aircraft Data

The Group 8 function `FlightDataAircraft8.p` was executed in MATLAB R2025b Update 7. Initial conditions were loaded from `ICsAircraft8_200knts_300ft.mat`. These files provide the aircraft parameter structure and initial vectors. The data do not identify a real aircraft type or manufacturer; the model is therefore referred to as the Group 8 aircraft.

2.1 Mass, Geometry and Propulsion

Table 1: Principal aircraft parameters extracted from the supplied data

Parameter	Description	Value
m	Mass	750 kg
g	Gravitational acceleration	9.81 m/s^2
S	Reference wing area	11.9 m^2
b	Wingspan	7 m
c	Reference chord	0.85 m
I_{xx}	Roll moment of inertia	605.29 kg m^2
I_{yy}	Pitch moment of inertia	1766.52 kg m^2
I_{zz}	Yaw moment of inertia	2168.83 kg m^2
I_{xz}	Product-of-inertia parameter	84.71 kg m^2
$P_{\max}$	Maximum power parameter	253.504 kW
η	Propeller efficiency parameter	0.85

The corresponding aspect ratio is $b^2/S = 4.11765$, and wing loading is $mg/S = 618.277\,\mathrm{N/m^2}$. The maximum power parameter is not the thrust at the trim condition.

The legacy fields are mapped to the names required by the current nonlinear simulation framework, as instructed in the data package:

```
FlightData = FlightDataAircraft8();
FlightData.Geometric = FlightData.Geo;
FlightData.Inertial = FlightData.I;
FlightData.Propeller = FlightData.Prop;
```

2.2 Initial State and Controls

$$\begin{aligned}\boldsymbol{X}_0 = [&102.8869153573,\ 0,\ 0.4724322704,\ 0,\ 0,\ 0,\\ &0,\ 0.004591730139,\ 0,\ 0,\ 0,\ -91.44]^{\mathsf{T}},\end{aligned} \tag{4}$$

Each entry uses the units specified in the state-vector definition above.

$$\boldsymbol{U}_0 = [0.7703166691,\ 0.01337052176,\ 0,\ 0,\ 0]^{\mathsf{T}}. \tag{5}$$

The principal conditions are also listed in Table 2 for clarity.

Table 2: Initial flight condition and controls

Quantity	Value
$V_0 = \sqrt{u_0^2 + v_0^2 + w_0^2}$	102.888 m/s, approximately 199.99827 knots
$h_0 = -Z_{e0}$	91.44 m, 300 ft
$\alpha_0 = \mathrm{atan2}(w_0, u_0)$	0.263086758°
θ_0	0.263086758°
β_0, ϕ_0, ψ_0	0
p_0, q_0, r_0	0
Throttle	77.0316669%
Elevator	+0.766074467°
Aileron, rudder and flap	0°

The body-axis velocity $w_0 = 0.4724322704$ m/s does not imply a climb. Earth-frame vertical speed is

$$V_s = u\sin\theta - v\sin\phi\cos\theta - w\cos\phi\cos\theta. \tag{6}$$

Substituting the Group 8 initial condition gives $V_{s0} = 0$, consistent with level flight and $\theta_0 = \alpha_0$.

2.3 Control Limits

The supplied data specify a throttle range of $[0, 1]$ and elevator, aileron and rudder limits of $\pm 25°$. The assignment explicitly limits the aileron to $\pm 20°$; the stricter assignment limit must be used for lateral performance assessment. No flap-deflection or control-rate limits are provided in these data.

The pitch-rate inner loop must respect the absolute elevator limit:

$$-25° \le \delta_{e0} + \Delta\delta_e(t) \le 25°. \tag{7}$$

Since $\delta_{e0} = 0.766074467°$, the available incremental elevator range is approximately

$$[-25.766074467°,\ 24.233925533°].$$

The remaining positive throttle increment is $1 - 0.7703166691 = 0.2296833309$.

3 Nonlinear Trim Verification

3.1 Method

With zero gust and the original $\boldsymbol{X}_0, \boldsymbol{U}_0$, the supplied `aero5560_aero.p` and `aero5560_motion.p` functions are evaluated. Because the aerodynamic model depends on state derivatives, the following consistency iteration is required:

$$(\boldsymbol{C}_F^{(k)}, \boldsymbol{C}_M^{(k)}) = \mathcal{A}(\boldsymbol{X}_0, \boldsymbol{0}, \dot{\boldsymbol{X}}^{(k)}, \boldsymbol{U}_0, F), \tag{8}$$

$$\dot{\boldsymbol{X}}^{(k+1)} = \mathcal{M}(\boldsymbol{X}_0, \boldsymbol{C}_F^{(k)}, \boldsymbol{C}_M^{(k)}, F). \tag{9}$$

The iteration terminates when the infinity norm of the update is below 10^{-11}, with a maximum iteration count enforced. At the initial condition, convergence is obtained in two iterations, with a final update norm of zero.

3.2 Residuals and Unforced Simulation

Table 3: Initial state-derivative residuals

Quantity	Result
$\dot{u}$	-2.4980×10^{-16} m/s^2
$\dot{v}, \dot{w}$	0 m/s^2
$\dot{p}, \dot{q}, \dot{r}$	0 rad/s^2
$\dot{\phi}, \dot{\theta}, \dot{\psi}$	0 rad/s
$\dot{X}_e$	102.888 m/s, expected forward motion
$\dot{Y}_e, \dot{Z}_e$	0 m/s

The diagnostic tolerances chosen for this check are 10^{-4} m/s^2 for translational acceleration, 10^{-6} in the corresponding SI units for angular acceleration and attitude rates, and 10^{-4} m/s for lateral and vertical earth-frame velocities. These are engineering checks rather than assignment grading thresholds; the actual residuals are much smaller.

With $\boldsymbol{U}_0$ fixed and no gust or feedback, the full nonlinear model is integrated for 60 s using `ode45`. The first run uses relative tolerance 10^{-9}, absolute tolerance 10^{-11} and maximum step 0.05 s. A tighter check uses 10^{-10}, 10^{-12} and 0.025 s. Outputs are sampled every 0.1 s, yielding 601 points.

Both runs retain the initial speed, altitude, vertical speed, angle of attack, pitch angle and three angular rates. Differences in these outputs are zero in the saved double-precision results. The final forward position is $X_e(60) \approx 6173.28$ m, consistent with $V_0 t$. The supplied trim point can therefore be retained for linearisation. Figure 1 shows the computed histories.

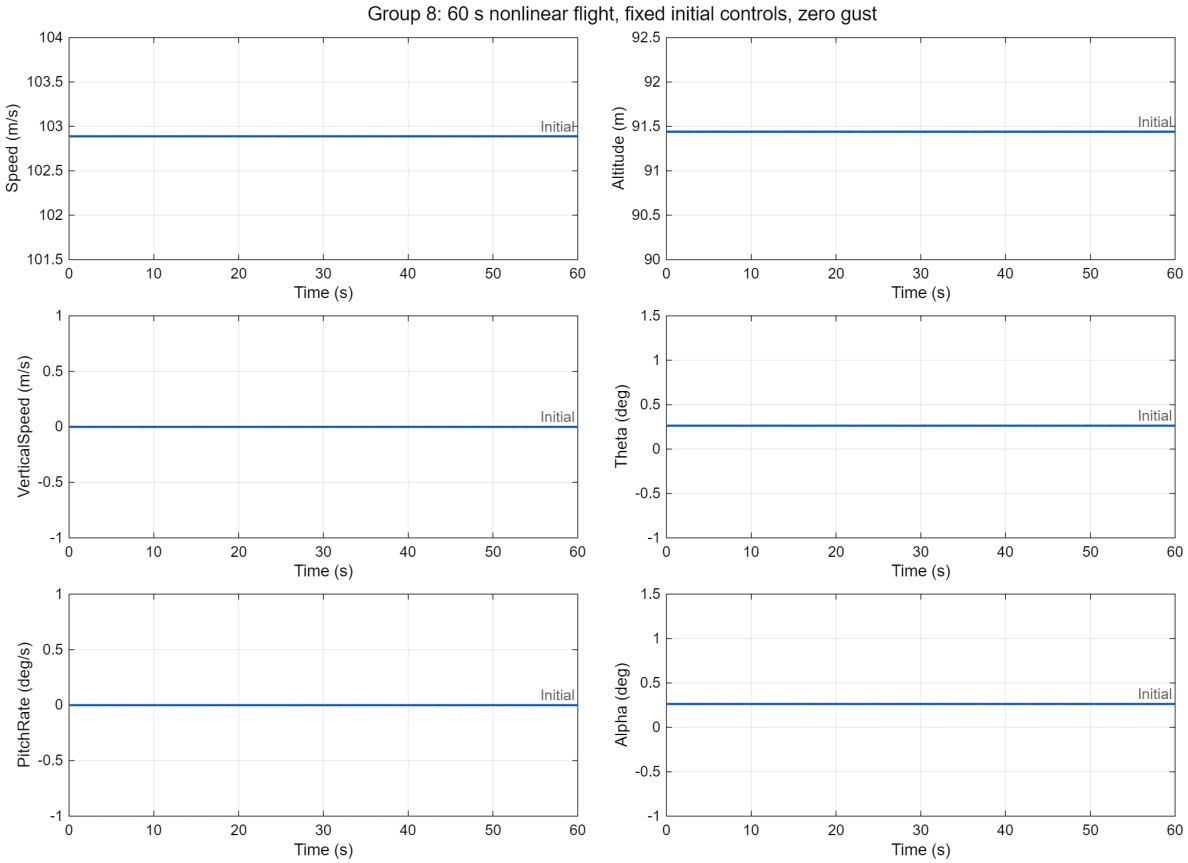


Figure 1: A 60 s nonlinear simulation of Group 8 with fixed initial controls and zero gust.

Remaining at an exact equilibrium without disturbance does not by itself prove disturbance stability. Stability must also be assessed using the linearised poles and perturbed responses.

4 Numerical Linearisation and State-Space Models

4.1 Central Differences and Numerical Checks

Near the nominal trim trajectory, the perturbation model is

$$\Delta \dot{\boldsymbol{X}} = A\Delta \boldsymbol{X} + B\Delta \boldsymbol{U}. \tag{10}$$

The Jacobian columns are calculated using central differences:

$$A_{:,i} \approx \frac{f(\boldsymbol{X}_0 + h_i \boldsymbol{e}_i, \boldsymbol{U}_0) - f(\boldsymbol{X}_0 - h_i \boldsymbol{e}_i, \boldsymbol{U}_0)}{2h_i}, \tag{11}$$

$$B_{:,j} \approx \frac{f(\boldsymbol{X}_0, \boldsymbol{U}_0 + k_j \boldsymbol{e}_j) - f(\boldsymbol{X}_0, \boldsymbol{U}_0 - k_j \boldsymbol{e}_j)}{2k_j}. \tag{12}$$

Every evaluation of f reconverges the implicit state derivatives. State increments are scaled according to velocity, angle/angular-rate and position magnitudes; control scales are unity. Relative steps of $10^{-4}, 3\times10^{-5}, 10^{-5}, 3\times10^{-6}, 10^{-6}$ are compared, and the middle value, 10^{-5}, is selected.

Table 4: Relative Frobenius-norm differences from the selected Jacobians

Relative step	$\|A_h - A\|_F / \|A\|_F$	$\|B_h - B\|_F / \|B\|_F$
10^{-4}	1.1803×10^{-9}	5.1592×10^{-12}
3×10^{-5}	9.5524×10^{-11}	5.5809×10^{-13}
10^{-5}	0	0
3×10^{-6}	1.1302×10^{-11}	1.6652×10^{-11}
10^{-6}	2.3360×10^{-10}	9.7214×10^{-11}

An independent mixed-perturbation Taylor check gives residuals of approximately 5.2498×10^{-5}, 1.3124×10^{-5} and 3.2808×10^{-6} as the perturbation amplitude is successively halved. The error ratios, 4.0002 and 4.0001, agree with the quadratic remainder expected from first-order linearisation. Agreement in a global norm does not imply identical relative accuracy for individual near-zero entries.

4.2 Four-State Longitudinal Model

Under the fixed-altitude approximation, state indices $[1, 3, 5, 8]$ and input indices $[1, 2]$ give

$$\boldsymbol{x}_\ell = [\Delta u, \Delta w, \Delta q, \Delta\theta]^\mathsf{T}, \qquad \boldsymbol{u}_\ell = [\Delta\delta_T, \Delta\delta_e]^\mathsf{T}. \tag{13}$$

$$A_\ell = \begin{bmatrix} -0.06144334 & 0.05552868 & -0.47243227 & -9.80989658 \\ -0.16960655 & -4.78731236 & 99.44823052 & -0.04504471 \\ 0.0002475794 & -0.11258862 & -3.67476184 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \tag{14}$$

$$B_\ell = \begin{bmatrix} 2.76318036 & 0 \\ -5.9699623 \times 10^{-5} & -62.90658081 \\ -1.7950804 \times 10^{-5} & -88.19693912 \\ 0 & 0 \end{bmatrix}. \tag{15}$$

The second state is body-axis velocity w, not angle of attack α. Changing to an angle-of-attack state requires a consistent transformation of both matrices.

The four-state longitudinal eigenvalues are

$$\lambda_{\mathrm{SP}} = -4.2313617 \pm 3.3004001\mathrm{i}, \qquad \lambda_{\mathrm{ph}} = -0.03039711 \pm 0.07729208\mathrm{i}. \tag{16}$$

These represent the short-period and phugoid modes, respectively, and lie in the left half-plane.

4.3 Altitude and Lateral-Directional Models

The Z_e column of the full A matrix contains nonzero terms. A five-state longitudinal model retaining altitude uses $[u, w, q, \theta, Z_e]$ and has also been saved. The four-state approximation omits altitude feedback. This report extracts $G_{q\delta_e}$ from Eqs. (14)–(15), consistently using the four-state model.

The lateral-directional states are $[v, p, r, \phi, \psi]$, with aileron and rudder inputs. Their poles are approximately -13.81679, $-0.719193 \pm 2.625197\mathrm{i}$, -0.0119723, and a zero heading-integrator pole. Position/heading zero eigenvalues in the full model do not imply divergent dynamics, but neither do they establish asymptotic return of all position states to the origin. Full and additional subsystem matrices are provided in the appendices.

4.4 Small-Step Linear and Nonlinear Comparisons

The full 12-state A, B model is compared with the nonlinear model for a $+0.05^\circ$ elevator increment and a $+0.001$ throttle increment over 10 s. The linear response is computed using matrix exponentials and the nonlinear response using `ode45`. Outputs are airspeed, q, and the vertical speed defined in Eq. (6).

Table 5: Maximum absolute output errors over the 10 s comparisons

Input step	Speed (m/s)	q (°/s)	V_s (m/s)
Elevator increment	0.016588	0.023974	0.641690
Throttle increment	2.0867×10^{-6}	1.3892×10^{-7}	2.5237×10^{-7}

For the elevator test, the maximum V_s error during the first 2 s is approximately 0.004968 m/s, or 0.2253% of the nonlinear peak over that interval. By 10 s, however, the vertical-speed peak is approximately 11.93 m/s and the speed change is approximately 4.69 m/s. The aircraft has departed appreciably from the initial point, and the relative V_s error grows to approximately 5.38%. The comparison supports local agreement; sustained-input motion of this magnitude should not be treated as an infinitesimal perturbation.

5 Elevator-to-Pitch-Rate Transfer Function

5.1 Output Selection and Derivation

The second longitudinal input column corresponds to the elevator. Set

$$\begin{aligned} B_q &= B_\ell(:,2) = \begin{bmatrix} 0 & -62.90658081 & -88.19693912 & 0 \end{bmatrix}^{\mathsf{T}}, \\ C_q &= \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D_q = 0. \end{aligned} \tag{17}$$

Taking the Laplace transform with zero initial perturbations gives

$$s\Delta \boldsymbol{x}_\ell = A_\ell \Delta \boldsymbol{x}_\ell + B_q \Delta \delta_e, \tag{18}$$

$$\Delta q = C_q \Delta \boldsymbol{x}_\ell, \tag{19}$$

and hence

$$\boxed{G_{q\delta_e}(s) = \frac{\Delta q(s)}{\Delta \delta_e(s)} = C_q (sI - A_\ell)^{-1} B_q.} \tag{20}$$

Here C_q denotes the state-space output matrix. The complementary sensitivity below is denoted $\mathcal{C}_q$ to avoid ambiguity.

5.2 ZPK and Polynomial Forms

The zero-pole-gain representation, with complex-conjugate pole factors combined into real quadratics, is

$$\boxed{G_{q\delta_e}(s) = \frac{-88.19693912\, s(s + 4.704978073)(s + 0.06347365466)}{(s^2 + 8.462723331 s + 28.79706207)(s^2 + 0.06079421306 s + 0.006898050455)}.} \tag{21}$$

The corresponding polynomial form is

$$G_{q\delta_e}(s) = \frac{-88.19693912 s^3 - 420.5628467 s^2 - 26.33932382 s}{s^4 + 8.523517544 s^3 + 29.31844472 s^2 + 1.809071019 s + 0.1986435871}. \tag{22}$$

The zeros are

$$z = 0, \quad -4.704978073, \quad -0.06347365466. \tag{23}$$

The poles are the longitudinal four-state eigenvalues given above, and the ZPK gain is -88.19693912.

5.3 Numerical Verification and Interpretation

Equation (21) is compared with the state-space frequency response at 501 points from 10^{-4} to 10^{4} rad/s. The maximum relative error is approximately 1.78×10^{-13}. An independently formed Cayley–Hamilton/Markov polynomial gives a maximum relative frequency-response error of approximately 7.20×10^{-9}.

One raw numerical zero is approximately $+2.40 \times 10^{-16}$. The model satisfies $q = \dot{\theta}$, the fourth entry of B_q is zero, and A_ℓ is nonsingular, so an exact zero at the origin is expected. After verifying this structure and the DC gain, only this floating-point residual is set to zero. The raw object is retained; no general pole-zero cancellation is applied.

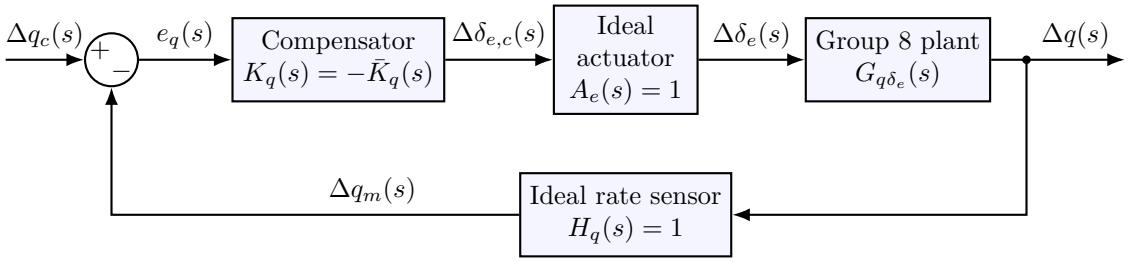
Since $G_{q\delta_e}(0) = 0$, pitch rate returns to zero following a constant elevator input in this stable open-loop model. The leading high-frequency term is approximately $-88.19694/s$, indicating a negative direct pitch-acceleration response to a positive elevator increment. This sign must be handled consistently in the feedback design.

6 Pitch-Rate Feedback-Loop Construction

6.1 Objective and Assumptions

The inner-loop objective is to make the actual pitch-rate increment Δq follow the command Δq_c using the elevator. Unity measurement, $H_q(s) = 1$, and an ideal actuator, $A_e(s) = 1$, are assumed for the linear perturbation design. Physical deflection limits are checked using Eq. (7) and enforced in the nonlinear implementation. Saturation is nonlinear and is not represented by a fixed LTI transfer function in this section.

The compensator is denoted $K_q(s)$. Its order and parameters will be selected through at least three subsequent design options. The block diagram below defines the architecture and all signals at this stage.



Rate signals: rad/s; elevator signals: rad; all signals are trim increments.

Figure 2: Unity negative-feedback pitch-rate loop using the elevator. The controller includes the required sign; its parameters remain to be designed.

Table 6: Signals and blocks in the pitch-rate feedback architecture

Symbol	Definition	Units
Δq_c	Commanded pitch-rate increment	rad/s
Δq_m	Measured feedback, equal to Δq for ideal measurement	rad/s
e_q	Command minus measured pitch rate	rad/s
$K_q(s)$	Signed compensator mapping pitch-rate error to elevator command	s
$\Delta\delta_{e,c}$	Commanded elevator-deflection increment	rad
$A_e(s) = 1$	Ideal actuator: actual increment equals command	Dimensionless
$\Delta\delta_e$	Actual elevator increment applied to the aircraft model	rad
$G_{q\delta_e}(s)$	Open-loop aircraft dynamics from elevator increment to q	(rad/s)/rad
Δq	Actual pitch-rate increment	rad/s
$H_q(s) = 1$	Unity pitch-rate measurement feedback	Dimensionless

6.2 Feedback Sign and Operation

The summing junction implements

$$e_q = \Delta q_c - \Delta q_m. \tag{24}$$

A command above the actual pitch rate produces positive error. In this aircraft model, a positive elevator increment has a negative direct effect on q, so the controller must generate an appropriate negative elevator action. The sign is retained within the compensator by writing $K_q = -\bar{K}_q$. This is a sign convention, not a stability proof. Root-locus, frequency-domain and closed-loop pole checks remain necessary; a minus sign at the summing junction alone does not establish stability.

The signal relationships are

$$\Delta\delta_{e,c} = K_q e_q, \tag{25}$$
$$\Delta\delta_e = A_e \Delta\delta_{e,c}, \tag{26}$$
$$\Delta q = G_{q\delta_e} \Delta\delta_e, \tag{27}$$
$$\Delta q_m = H_q \Delta q. \tag{28}$$

For example, a 4°/s command corresponds to $\Delta q_c = 4\pi/180 \approx 0.06981317$ rad/s. The comparator, controller and aircraft model all use radians internally to avoid an unintended $180/\pi$ scaling factor.

6.3 Closed-Loop Derivation and SCK Expressions

Define the signed loop transfer function as

$$L_q = K_q A_e G_{q\delta_e} H_q. \tag{29}$$

Combining Eq. (24) with the forward-path relationships gives

$$\Delta q = G_{q\delta_e} A_e K_q (\Delta q_c - H_q \Delta q), \tag{30}$$

Thus the general closed-loop relationships are

$$\frac{\Delta q}{\Delta q_c} = \frac{G_{q\delta_e} A_e K_q}{1 + L_q}, \qquad \frac{\Delta\delta_e}{\Delta q_c} = \frac{A_e K_q}{1 + L_q}. \tag{31}$$

For the assumptions $A_e = H_q = 1$, define sensitivity and complementary sensitivity by

$$S_q = \frac{1}{1 + G_{q\delta_e} K_q}, \qquad \mathcal{C}_q = \frac{G_{q\delta_e} K_q}{1 + G_{q\delta_e} K_q}, \qquad S_q + \mathcal{C}_q = 1. \tag{32}$$

The resulting SCK relationships are

$$\boxed{\frac{\Delta q}{\Delta q_c} = \mathcal{C}_q, \quad \frac{e_q}{\Delta q_c} = S_q, \quad \frac{\Delta \delta_e}{\Delta q_c} = K_q S_q.} \tag{33}$$

The complementary sensitivity $\mathcal{C}_q$ describes command tracking, S_q describes tracking error and responses at appropriate disturbance locations, and $K_q S_q$ gives elevator demand. No noise or gust input is silently introduced at an unspecified point. Subsequent gust transfer functions must be derived for their actual injection locations.

Let N_G, D_G be the numerator and denominator in Eq. (21), and write $K_q = N_K / D_K$. The closed-loop characteristic equation is

$$D_G D_K + N_G N_K = 0. \tag{34}$$

The numerator N_K already includes the controller sign. When using $K_q = -\bar{K}_q$, the minus sign must not be duplicated or omitted elsewhere. Once the compensator is selected, these relationships provide the numerical closed-loop ZPK forms. Symbolic relationships are retained at this stage.

6.4 Trim Bias and MATLAB Construction

The linear loop produces the incremental elevator signal $\Delta\delta_e$. For the nonlinear aircraft, the absolute input is

$$\delta_e(t) = 0.01337052176 + \Delta\delta_e(t) \quad \text{(rad)}, \tag{35}$$

and must satisfy the absolute deflection limits. The ideal actuator in the linear diagram does not automatically implement saturation.

Once K_q has been supplied by the subsequent design process, the same feedback structure can be constructed in MATLAB:

```
load('Gqde.mat', 'Gqde');
% Kq must contain the intended controller sign.
Lq = Gqde * Kq;
Sq = feedback(1, Lq); % error / q_command
Cq_closed = feedback(Lq,1); % q / q_command
Uq = feedback(Kq,Gqde); % elevator_increment / q_command
% q_command = deg2rad(4); % subsequent performance test
```

An equivalent function is provided in `loop.m`. The caller must supply the selected, signed K_q; the function does not choose an arbitrary controller or imply that performance specifications are met.

7 Specifications for Subsequent Compensator Design

The next stage compares at least three compensator designs using a 4°/s pitch-rate command. The assignment specifies a settling time of 3 s, steady-state error below 1%, maximum overshoot below 20%, and elevator deflection within its limits. It also gives a 2–10 rad/s frequency target and a noise bandwidth of 20 rad/s.

The assignment associates crossover frequency with a −3 dB point. Open-loop 0 dB gain crossover and closed-loop −3 dB bandwidth are generally different quantities and should be reported separately. The rise-time convention in Lecture 8-2 also differs from MATLAB's common 10–90% default; the chosen definition must be stated.

Each design should document the compensator ZPK form, root locus, Bode plot, step response, elevator activity and a specification comparison table. The plant and feedback architecture are established here, but closed-loop compliance has not yet been demonstrated.

8 Files, Reproduction and Sources

Table 7: Current project scripts and outputs

Directory or entry point	Purpose
`trim/run_trim.m`	Trim residuals, 60 s unforced simulation and integration check
`linear/run_linear.m`	Central-difference A/B, subsystems, eigenvalues and local validation
`Gqde/run_Gqde.m`	Extraction of the Group 8 $G_{q\delta_e}$ and frequency-response verification
`AI_draft/loop.m`	Feedback transfer relationships for a supplied signed compensator
`AI_draft/report_en.tex`	Single English LaTeX source, including a native TikZ block diagram
`AI_draft/trim.png`	Figure exported from the actual trim simulation

Printed matrices and coefficients are rounded. Further numerical work should use the full-precision MAT/JSON files. This English backup preserves the technical content of the Chinese progress report and can support the modelling and inner-loop architecture sections of the final assignment report.

References

[1] AERO5560, *Group Project: Aircraft Closed-Loop Control System Design and Analysis*, assignment brief, pp. 1–5.

[2] *Group8 AircraftData*: `FlightDataAircraft8.p`, `ICsAircraft8_200knts_300ft.mat`, and `Note.txt`.

[3] AERO5560, *Flight Simulation Student Version*, supplied nonlinear functions and interface documentation.

[4] AERO5560, Lectures 1-2, 2-1/2, 7-2, 8-1/2, 9-1/2 and 10-1: linearisation, transfer functions, feedback architectures and controller design.

A Full Twelve-State Matrices

State and input ordering follows Section 1. The matrices below retain approximately eight significant figures. Each element has units given by its row's state-derivative units divided by its column-variable units.

$$A = \begin{bmatrix}
-0.061443341 & 0 & 0.05552868 & 0 & -0.47243227 & 0 & 0 & -9.8098966 & 0 & 0 & 0 & 3.1754685 \times 10^{-5} \\
0 & -0.37729298 & 0 & 0.53108604 & 0 & -102.86881 & 9.8487485 & 0 & 0 & 0 & 0 & 0 \\
-0.16960655 & 0 & -4.7873124 & 0 & 99.448231 & 0 & 0 & -0.045044714 & 0 & 0 & 0 & -0.00094886847 \\
0 & -0.20268929 & 0 & -13.808594 & 0 & 1.4524332 & 0.019116101 & 0 & 0 & 0 & 0 & 0 \\
0.00024757942 & 0 & -0.11258862 & 0 & -3.6747618 & 0 & 0 & 0 & 0 & 0 & 0 & -1.3364074 \times 10^{-6} \\
0 & 0.058710969 & 0 & -0.54309133 & 0 & -1.0812624 & -0.0033528581 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0.0045917624 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1.0000105 & 0 & 0 & 0 & 0 & 0 & 0 \\
0.99998946 & 0 & 0.004591714 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & -0.47243227 & 0 & 102.888 & 0 & 0 & 0 \\
-0.004591714 & 0 & 0.99998946 & 0 & 0 & 0 & 0 & -102.888 & 0 & 0 & 0 & 0
\end{bmatrix}$$

A.1 Full Input Matrix

$$B = \begin{bmatrix} 2.7631804 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3.7014393 & 20.777334 & 0 \\ -5.9699623 \times 10^{-5} & -62.906581 & 0 & 0 & 0 \\ 0 & 0 & -318.33488 & -3.5101752 & 0 \\ -1.7950804 \times 10^{-5} & -88.196939 & 0 & 0 & 0 \\ 0 & 0 & -16.168938 & -27.867108 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{36}$$

B Additional Subsystem Matrices

B.1 Five-State Longitudinal Model with Altitude

States: $[u, w, q, \theta, Z_e]$; inputs: $[\delta_T, \delta_e]$.

$$A_{\ell,5} = \begin{bmatrix} -0.061443341 & 0.05552868 & -0.47243227 & -9.8098966 & 3.1754685 \times 10^{-5} \\ -0.16960655 & -4.7873124 & 99.448231 & -0.045044714 & -0.00094886847 \\ 0.00024757942 & -0.11258862 & -3.6747618 & 0 & -1.3364074 \times 10^{-6} \\ 0 & 0 & 1 & 0 & 0 \\ -0.004591714 & 0.99998946 & 0 & -102.888 & 0 \end{bmatrix} \tag{37}$$

$$B_{\ell,5} = \begin{bmatrix} 2.7631804 & 0 \\ -5.9699623 \times 10^{-5} & -62.906581 \\ -1.7950804 \times 10^{-5} & -88.196939 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{38}$$

B.2 Five-State Lateral-Directional Model

States: $[v, p, r, \phi, \psi]$; inputs: $[\delta_a, \delta_r]$.

$$A_{\text{lat}} = \begin{bmatrix} -0.37729298 & 0.53108604 & -102.86881 & 9.8487485 & 0 \\ -0.20268929 & -13.808594 & 1.4524332 & 0.019116101 & 0 \\ 0.058710969 & -0.54309133 & -1.0812624 & -0.0033528581 & 0 \\ 0 & 1 & 0.0045917624 & 0 & 0 \\ 0 & 0 & 1.0000105 & 0 & 0 \end{bmatrix} \tag{39}$$

$$B_{\text{lat}} = \begin{bmatrix} 3.7014393 & 20.777334 \\ -318.33488 & -3.5101752 \\ -16.168938 & -27.867108 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{40}$$

C Complete Extracted Aerodynamic Parameters

Original parameter-field names are retained. Angle, angular-rate and rate-of-change derivatives may use nondimensional variables within the aerodynamic model and must not be inserted directly as dimensional A/B entries.

Table 8: Original Group 8 aerodynamic fields

Field	Value	Field	Value
alpha_o	−0.013772917	Cmad	0.9531
Cdo	0.020342	Cmde	−2.3941614
k	0.063776	Cnb	0.027705
CLa	4.786	Cnbd	−0.0051
CLq	9.269	Cnp	−0.000421
CLad	−1.1439	Cnr	−0.140618
CLde	0.61398157	Cnda	−0.015126086
CLo	0.065917179	Cndr	−0.11229973
Cyb	−0.379162	Clb	−0.024446
Cybd	0.117	Clbd	0.0068
Cyp	0.016304	Clp	−0.456441
Cyr	0.1222	Clr	0.0601
Cyda	0.036153637	Clda	−0.35735378
Cydr	0.20294165	Cldr	0.000394
Cmo	0.03337	CDdf	0
Cma	−0.295926	CLdf	0
Cmq	−25.094866	Cmdf	0